

CCNA 200-301 V2.0

17 OSPFv2, Single Area

Part III — IP Routing

EXAM TOPICS

3.3

LECTURER

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Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

What you will be able to do

- **Explain** how OSPF forms adjacencies and why they sometimes do not.
- **Configure** single-area OSPFv2 with an explicit router ID, on both broadcast and point-to-point links.
- **Predict** and **influence** DR and BDR election.
- **Troubleshoot** the five mismatches that prevent an adjacency, using `show ip ospf neighbor` and `show ip ospf interface`.

Before we start

Chapter 15 for administrative distance and metrics; Chapter 16 for the contrast with a route you configure by hand.

Vocabulary for this chapter

OSPF

link-state

adjacency

neighbour

hello

router ID

area 0

DR

BDR

DROTHER

cost

reference bandwidth

wildcard mask

passive interface

FULL state

2-WAY state

Each is defined where it first matters — not here.

What the blueprint asks for, precisely

This is what the blueprint actually asks for

Topic **3.3** is “configure single area OSPFv2 for IPv4 and OSPFv3 for IPv6” with four sub-points: neighbour adjacencies *excluding authentication*, point-to-point, broadcast with DR/BDR selection, and router ID. Multi-area OSPF, authentication, virtual links and route summarisation are all out of scope. Learn these four properly rather than the whole protocol thinly.

What OSPF does that a static route cannot

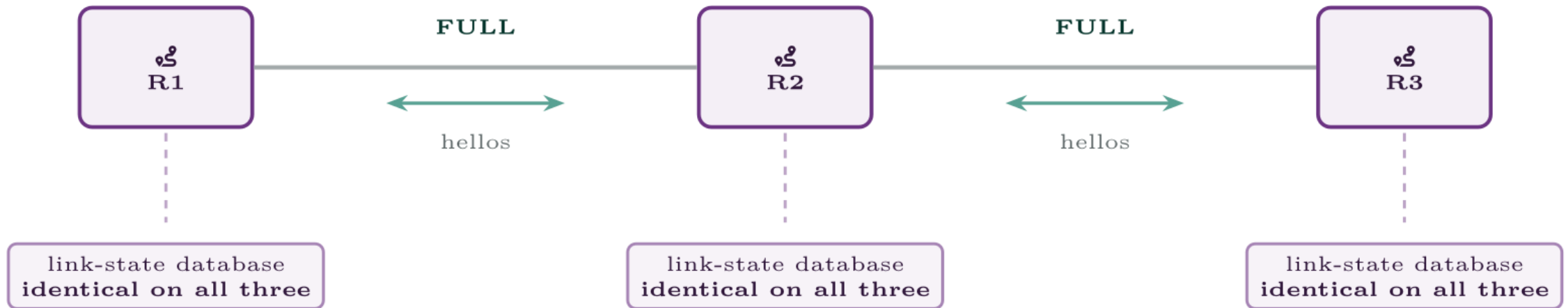


Link state, in three sentences

Every OSPF router floods a description of its own links to every other router in the area. Each router therefore ends up with an identical map of the whole area, which it stores in a **link-state database**. Each then runs the shortest-path algorithm against that map to work out its own best route to everywhere.

Because every router has the whole map, a failure anywhere is known everywhere within seconds, and every router recalculates independently and consistently. That is the property a static route can never have.

What OSPF does that a static route cannot



Each router floods a description of *its own* links. Every router therefore holds the same map — and then computes its *own* best path across it. Nobody is told what their routes are.

Link state in one picture. The database is shared and identical; the routing table each router builds from it is its own.

Router ID, and how it is chosen

The **router ID** is a 32-bit value written like an IP address that uniquely identifies the router in OSPF. It is selected in this order:

- The value configured with `router-id` — always use this.
- Otherwise, the highest IP address on any *loopback* interface.
- Otherwise, the highest IP address on any active physical interface.

Left to itself the ID can change when an interface goes down, which forces the adjacency to rebuild. Configure it explicitly and the problem disappears.

Configuring single-area OSPFv2



OSPF on R1, area 0

IOS · configuration mode

```
! A loopback gives a stable ID and a pingable address that never  
! goes down with a physical interface.  
interface Loopback0  
  ip address 1.1.1.1 255.255.255.255  
!  
router ospf 10  
  ! The process number is LOCAL. R2 may use a different one and  
  ! the two will still peer -- unlike EIGRP or BGP.  
  router-id 1.1.1.1  
  !  
  ! The network statement uses a WILDCARD mask -- the inverse of a  
  ! subnet mask. It means "enable OSPF on any interface whose  
  ! address falls in this range, and advertise that subnet".  
  network 10.255.0.0 0.0.0.3 area 0  
  network 192.168.10.0 0.0.0.255 area 0  
  !  
  ! Do not send hellos out of a LAN with no other routers on it.  
  ! It saves bandwidth and removes an attack surface, while the  
  ! subnet is still advertised.  
  passive-interface Vlan10  
  !
```

OSPF on R1, area 0 (continued)

IOS · configuration mode

```
! Make cost reflect modern link speeds -- see the note below.  
auto-cost reference-bandwidth 10000
```

Wildcard masks, quickly

A wildcard mask is the subnet mask with every bit inverted: `0` means “this bit must match”, `255` means “do not care”.

- `/24` → mask `255.255.255.0` → wildcard `0.0.0.255`
- `/30` → mask `255.255.255.252` → wildcard `0.0.0.3`
- A single interface → wildcard `0.0.0.0`

The arithmetic is 255 minus each octet of the subnet mask. Wildcards return in Chapter 24.

Reference bandwidth: change it everywhere or nowhere

OSPF cost is $\text{reference bandwidth} \div \text{interface bandwidth}$, and the default reference is 100 Mbps. That means a 1 Gbps link and a 10 Gbps link both come out at cost 1 — OSPF cannot tell them apart, and will happily choose the slower one.

Raising the reference to 10000 (10 Gbps) restores the distinction. But cost is compared across the area, so **every router must use the same reference bandwidth** or the path calculations disagree and traffic takes strange routes. Change it on all of them, or on none.

DR and BDR on broadcast links



Why a designated router exists

On a broadcast segment with n routers, full adjacencies between every pair would be $n(n-1)/2$ relationships — ten routers would need forty-five. Instead one router is elected **DR** and one **BDR**, and everyone else becomes a **DROTHER** adjacent only to those two. Flooding goes through the DR.

The election: **highest OSPF interface priority** wins; if priorities tie, **highest router ID** wins. Default priority is 1. Priority 0 means “never be DR”.

Two things surprise people. The election is **per segment**, not per router — a router can be DR on one link and DROTHER on another. And it is **non-preemptive**: a new router with a better priority does not take over until the current DR fails. To force it, clear the process or bounce the interface.

Influencing the election, and avoiding it

IOS · configuration mode

```
! Make this router the DR on the segment.
interface GigabitEthernet0/0
  ip ospf priority 255
!
! Never allow this one to be DR or BDR.
interface GigabitEthernet0/1
  ip ospf priority 0
!
! On a link with exactly two routers, skip the election entirely.
! Point-to-point forms an adjacency faster and elects nobody.
interface GigabitEthernet0/2
  ip ospf network point-to-point
```

Verifying



show ip ospf neighbor

R1#

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:33	10.255.0.2	GigabitEthernet0/1
3.3.3.3	1	FULL/BDR	00:00:31	10.255.0.3	GigabitEthernet0/1
4.4.4.4	0	FULL/ -	00:00:38	10.255.1.2	GigabitEthernet0/2

show ip ospf interface GigabitEthernet0/1

R1#

```
GigabitEthernet0/1 is up, line protocol is up
  Internet Address 10.255.0.1/30, Area 0, Attached via Network Statement
  Process ID 10, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10
  Designated Router (ID) 2.2.2.2, Interface address 10.255.0.2
  Backup Designated router (ID) 3.3.3.3, Interface address 10.255.0.3
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Neighbor Count is 2, Adjacent neighbor count is 2
```

Common pitfalls

- **Assuming the process number must match.** It is locally significant. `router ospf 10` peers happily with `router ospf 65`.
- **Using a subnet mask in the network statement.** It takes a wildcard. `network 10.255.0.0 255.255.255.252` does not do what you expect.
- **Leaving the router ID to chance.** It changes when interfaces change, and every adjacency rebuilds.
- **Panicking at 2-WAY.** Between two DROTHERs it is correct.
- **Changing reference bandwidth on one router.** See the warning above.

Common pitfalls (continued)

- **Forgetting `passive-interface` on user VLANs.** Hellos on a LAN full of PCs are wasted and expose the protocol.

Adjacency stuck in EXSTART.

Two routers see each other but never reach FULL:

Hellos are being exchanged — so area, timers, subnet and network type all match, or the neighbour would not be listed at all. The failure is later, during database exchange.

Adjacency stuck in EXSTART.

What would you check first — and what do you expect to see?

show ip ospf neighbor

R1#

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	EXSTART/DR	00:00:35	10.255.0.2	GigabitEthernet0/1

show ip ospf interface GigabitEthernet0/1 | include MTU|Internet

R1#

```
Internet Address 10.255.0.1/30, Area 0, Attached via Network Statement  
MTU is 1500 bytes
```

show ip ospf interface GigabitEthernet0/1 | include MTU|Internet

R2#

```
Internet Address 10.255.0.2/30, Area 0, Attached via Network Statement  
MTU is 1400 bytes
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. An MTU mismatch. OSPF checks MTU during database description exchange and refuses to proceed if the two ends disagree — which is exactly why the adjacency reaches EXSTART and stops there. Being stuck in EXSTART or EXCHANGE is close to diagnostic of this.

And the lesson

Fix. Make the MTUs match: `ip mtu 1500` on R2's interface, or 1400 on R1's if the smaller value was deliberate. The adjacency will move to FULL within a dead interval. Do not use `ip ospf mtu-ignore` except as a temporary measure — it suppresses the symptom and leaves a real MTU mismatch to fragment traffic later.

Three routers, one area

Topology. R1, R2, R3 on a shared Ethernet segment; each with a LAN; R1 also point-to-point to R3.

1. Configure loopbacks and explicit router IDs 1.1.1.1, 2.2.2.2, 3.3.3.3. Enable OSPF area 0 on all links and LANs, with LANs passive.
2. Confirm every neighbour reaches FULL. Record which router became DR and which BDR, and explain the election from the router IDs.
3. Set `ip ospf priority 255` on R3's segment interface. Note that nothing changes, then bounce the interfaces and note that it does. Explain why.
4. On the R1–R3 link, set `ip ospf network point-to-point` at both ends. Confirm the state becomes FULL/ - with no DR.

Three routers, one area (continued)

5. Change the hello interval on **R1** only. Watch the adjacency drop, and find the mismatch with `show ip ospf interface`.
6. Set `ip mtu 1400` on one end and reproduce the EXSTART fault above.
7. **Extension.** Raise the reference bandwidth on one router only and observe the asymmetric routing that results.

CHECKPOINT 1 of 3

Two routers on a broadcast segment show 2-WAY and never go FULL. Under what circumstance is this correct rather than a fault?

Discuss · then advance

Checkpoint 1 of 3

Two routers on a broadcast segment show 2-WAY and never go FULL. Under what circumstance is this correct rather than a fault?

When both routers are DROTHERs. Two non-designated routers on a broadcast segment deliberately stop at 2-WAY — they exchange databases with the DR and BDR, not with each other. It is a fault only if one of them should have been the DR or BDR.

CHECKPOINT 2 of 3

Which wildcard mask enables OSPF on 172.16.8.0/22?

Discuss · then advance

Checkpoint 2 of 3

Which wildcard mask enables OSPF on 172.16.8.0/22?

0.0.3.255. The mask for /22 is 255.255.252.0, and $255 - 252 = 3$.

CHECKPOINT 3 of 3

A router's OSPF adjacencies all rebuild every time one interface flaps. What is the cause, and what is the one-line fix?

Discuss · then advance

Checkpoint 3 of 3

A router's OSPF adjacencies all rebuild every time one interface flaps. What is the cause, and what is the one-line fix?

The router ID was derived from that interface's address, so when the interface flaps the router ID changes and every adjacency rebuilds. Fix it with an explicit `router-id`, or a loopback, which never flaps.

What to take away

- OSPF floods a description of every link, so all routers in the area share one map and converge consistently.
- Area, hello and dead timers, subnet and mask, and network type must all match, or no adjacency forms. Router IDs must be unique.
- Configure the router ID explicitly; otherwise it moves and adjacencies rebuild.
- Network statements take wildcard masks. The process number is local.
- DR and BDR are elected per segment by priority then router ID, and the election is non-preemptive. `ip ospf network point-to-point` avoids it entirely on two-router links.
- **FULL** is healthy; **2-WAY** between DROTHERs is healthy; stuck in **EXSTART** means MTU.

Where we got to

- OSPF floods a description of every link, so all routers in the area share one map and converge consistently.
- Area, hello and dead timers, subnet and mask, and network type must all match, or no adjacency forms. Router IDs must be unique.
- Configure the router ID explicitly; otherwise it moves and adjacencies rebuild.
- Network statements take wildcard masks. The process number is local.

NEXT

Chapter 18 — OSPFv3 for IPv6